

EXP 25: Prolog Program to Implement Monkey Banana Problem

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Aim:

To develop a Prolog program that solves the classic Monkey Banana problem, so that the program determines whether the monkey can obtain the banana using a sequence of valid actions.

Algorithm:

Step 1: Start the program.

Step 2: Represent the positions of the monkey, box, and banana as a state.

Step 3: Check whether the monkey already possesses the banana.

Step 4: If not, determine a valid action such as walking, pushing the box, climbing the box, or grasping the banana.

Step 5: Apply the selected action to generate a new state.

Step 6: Verify whether the new state leads closer to obtaining the banana.

Step 7: Continue exploring valid actions recursively.

Step 8: Display whether the monkey can successfully obtain the banana.

Step 9: Stop the program.

Source Code:

```
% Program to implement the Monkey Banana Problem
```

```
move(state(middle, onbox, middle, hasnot), grasp,  
    state(middle, onbox, middle, has)).
```

```
move(state(P, onfloor, P, H), climb,  
    state(P, onbox, P, H)).
```

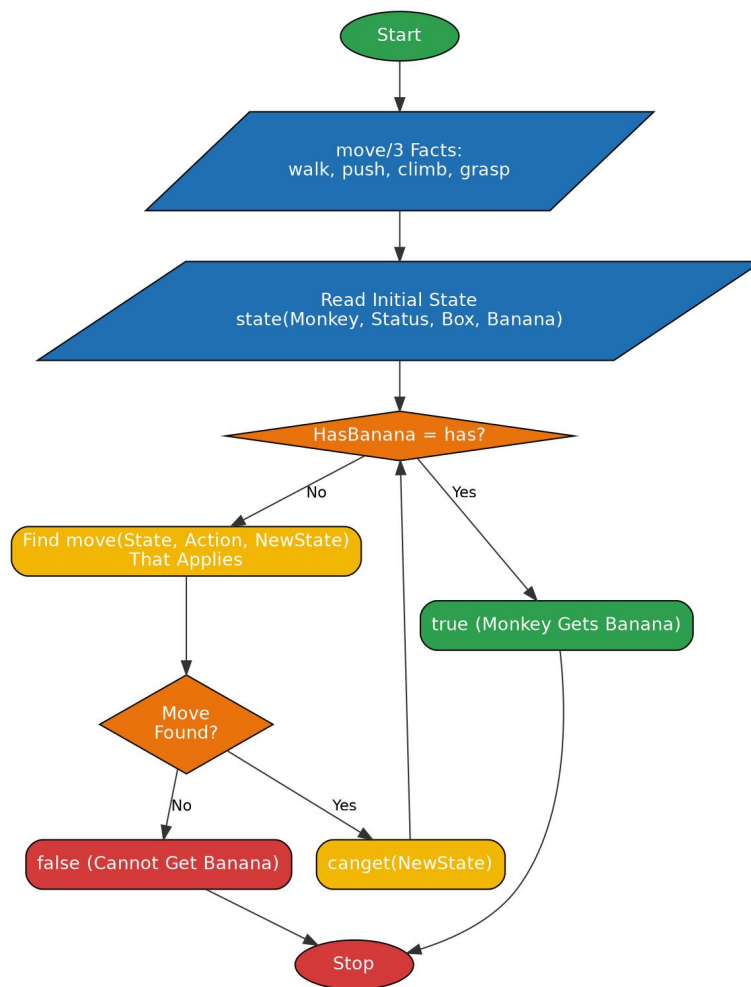
```
move(state(P1, onfloor, P1, H), push(P1, P2),  
    state(P2, onfloor, P2, H)).
```

```
move(state(P1, onfloor, B, H), walk(P1, P2),  
    state(P2, onfloor, B, H)).
```

```
canget(state(_, _, _, has)) :- !.
```

```
canget(State1) :-  
    move(State1, _, State2),  
    canget(State2).
```

Flowchart:



Output:

```
Welcome to SWI-Prolog (threaded, 64 bits, version 9.2.9)
SWI-Prolog comes with ABSOLUTELY NO WARRANTY. This is free software.
Please run ?- license. for legal details.

For online help and background, visit https://www.swi-prolog.org
For built-in help, use ?- help(Topic). or ?- apropos(Word).

?- cd('C:/Users/yanar/Documents/CSA1717').
true.

?- [monkey].
true.

?- canget(state(atdoor, onfloor, atwindow, hasnot)).
true.

?- canget(state(middle, onbox, middle, hasnot)).
true.
```

Result:

Thus the Prolog program to implement the Monkey Banana problem was written, executed successfully and the output was verified.